

■ Composting Guide

Comprehensive Environmental Guide

■ **Quick Tip:** Maintain a 3:1 ratio of browns (carbon) to greens (nitrogen) — this is the golden rule for fast, odour-free composting.

■ Definition

Composting is a controlled biological process in which micro-organisms — bacteria, fungi, and other decomposers — break down organic matter such as food scraps, garden waste, and paper into a nutrient-rich, humus-like material called compost. This natural recycling process mirrors the decomposition that occurs on forest floors, but is accelerated and managed to produce a stable, beneficial soil amendment.

Composting diverts organic waste from landfills, where it would otherwise generate methane — a greenhouse gas more than 80 times more potent than CO₂ over a 20-year period. The resulting compost improves soil structure, water retention, and microbial diversity, reducing the need for synthetic fertilisers.

■■ Disposal Methods

- Backyard composting: an outdoor pile or bin fed with kitchen scraps and garden trimmings, ideal for households with outdoor space.
- Vermicomposting (worm bins): red wigglers convert organic waste into rich worm castings; suitable for indoor use in small spaces.
- Bokashi fermentation: an anaerobic system using inoculated bran to ferment all food waste, including meat and dairy.
- Municipal/community composting: kerbside organic waste collection processed at industrial-scale in-vessel or windrow facilities.
- Tumbler composting: enclosed rotating drums that speed decomposition and deter pests.

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■ Environmental Impact

Organic waste in anaerobic landfill conditions produces methane, a potent greenhouse gas contributing significantly to climate change. Composting diverts this waste into an aerobic process that produces far less warming potential. Globally, food waste accounts for approximately 8–10% of total greenhouse gas emissions; widespread composting could eliminate a significant fraction of this.

Applying compost to agricultural land sequesters carbon in the soil, improves water infiltration (reducing run-off and flooding risk), and suppresses plant diseases — reducing dependence on chemical pesticides. Healthy compost-amended soils also support greater biodiversity of soil organisms fundamental to ecosystem function.

■■ Safety Guidelines

- Do not compost meat, fish, dairy, or oily foods in open bins — they attract rodents and produce odour; use a sealed system like bokashi instead.
- Avoid composting diseased plants or weeds with mature seeds, as pathogens and seeds may survive.
- Do not add pet faeces, treated wood, or synthetic materials to compost.
- Turn the pile regularly to maintain aerobic conditions and prevent anaerobic pockets that produce foul smells.
- Ensure the pile reaches 55–65°C in the active phase to kill weed seeds and pathogens; use a compost thermometer to monitor.

■ Best Practices

- Balance 'greens' (nitrogen-rich: fruit peels, grass clippings, coffee grounds) with 'browns' (carbon-rich: dry leaves, cardboard, straw) in roughly a 1:3 ratio by volume.
- Keep the pile moist — about as damp as a wrung-out sponge — to sustain microbial activity.
- Chop or shred materials into smaller pieces to accelerate decomposition.
- Site the compost heap in partial shade to regulate moisture and temperature.
- Use finished compost as a mulch, soil conditioner, or potting mix amendment to close the nutrient cycle.